

Data-Driven 3D Garment Deformation from Human Pose

UE24CS352A - Machine Learning Mini-Project Final Review & Demonstration

Reference Study: Xue & Wu (Stanford CS 229) | **Alternative Dataset:** TailorNet (CVPR 2020)

Team: 2-Member Student Team

1. Problem Statement & Motivation

The Real-Time Cloth Challenge:

- Interactive 3D applications (gaming, VR, character animation, and virtual try-on) demand realistic clothing behavior.
- **The Bottleneck:** Traditional physics-based cloth simulation (PDE solvers like PhysBAM using mass-spring/finite element models) requires solving complex time-stepping and contact equations, taking **~180 seconds per frame**.
- **The Solution:** A data-driven machine learning model that maps human body pose parameters directly to 3D garment mesh displacements.
- **Our Goal:** Reproduce the core methodology of the reference paper using the public TailorNet dataset, and introduce architectural improvements that outperform the paper's baseline in accuracy and stability.

2. Original Paper & Dataset Adaptation

Original Paper (Xue & Wu, 2021):

- Focused on predicting loose coat deformation from 14 skeletal joints (56 quaternion dimensions).
- Used proprietary synthetic PhysBAM coat simulation data (9,813 poses). *Dataset is not publicly available.*

Public Alternative: TailorNet Dataset (CVPR 2020, MPI):

- TailorNet provides physically simulated 3D garment displacements across SMPL body poses.
- We utilized the curated t-shirt_female dataset (110 frames, 7,702 vertices, 15,180 triangular faces).
- **Pose Input:** Converted 24 SMPL joint rotations into 96-dimensional unit quaternions (w, x, y, z) to prevent gimbal lock.
- **Deformation Output:** 3D vertex displacements from canonical t-shirt template ($7,702 \text{ times } 3 = 23,106 \text{ dims}$).

3. PCA Dimensionality Reduction & Analysis

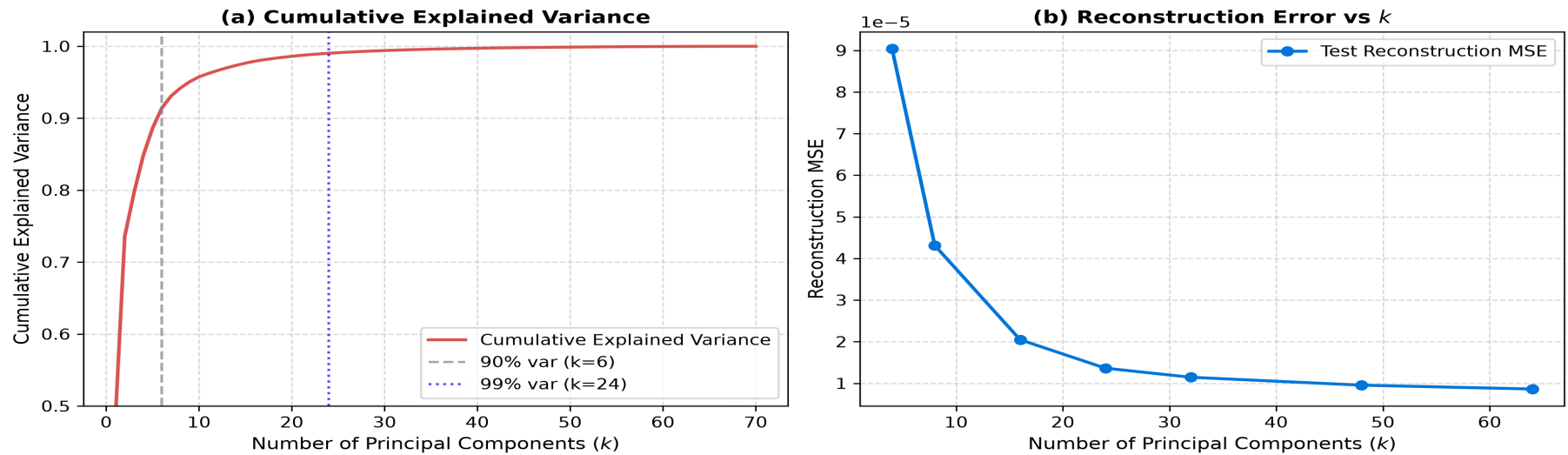


Figure 1: (a) Cumulative explained variance vs principal components. (b) Test reconstruction MSE vs k .

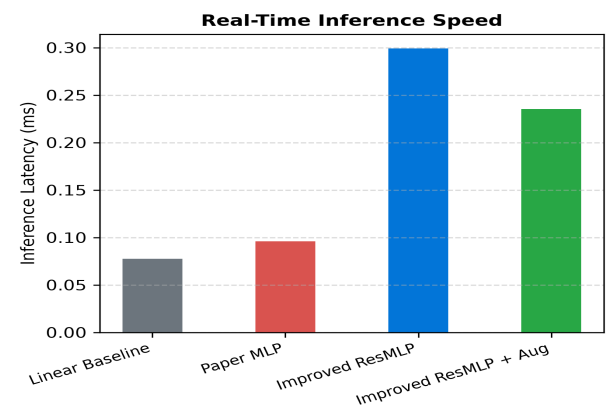
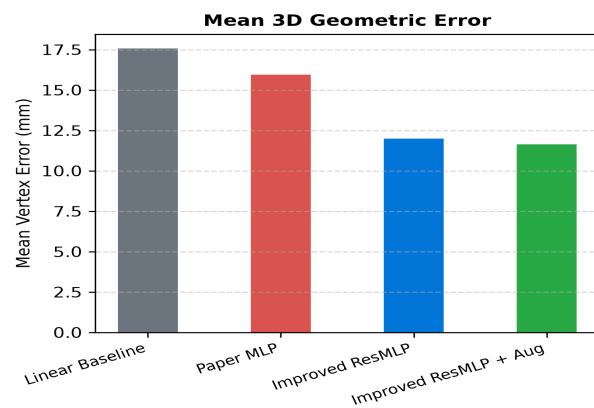
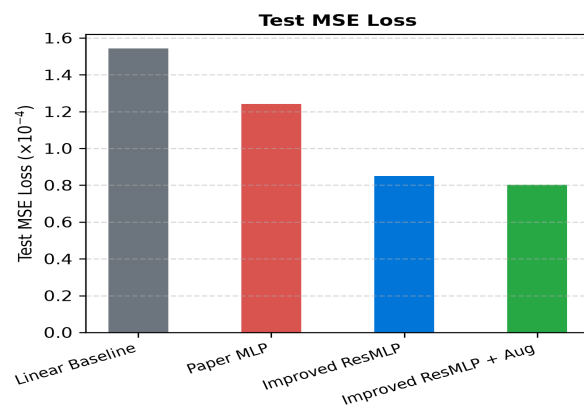
- Fitted strictly on the 70 training samples: $k=32$ principal components captures **99.49% of total variance** with test reconstruction MSE of 1.15×10^{-5} .

4. Model Architecture & Proposed Improvements

1. **Linear Baseline (Paper ID 0):** Linear layer (96 $\rightarrow$ 32) + Fixed PC Layer.
2. **Paper MLP Baseline (Paper ID 3):** Linear(96,128) $\rightarrow$ ReLU $\rightarrow$ Linear(128,256) $\rightarrow$ ReLU $\rightarrow$ Linear(256,32) $\rightarrow$ PC Layer.
3. **Proposed Improvements (ResMLP + Augmentation):**
 - **Layer Normalization:** Stabilizes hidden activations across poses and prevents internal covariate shift.
 - **GELU Activations:** Continuous non-linear gating that prevents dead neurons on quaternion coordinates.
 - **Residual Skip Connections:** Direct identity paths ($h_{l+1} = h_l + \mathcal{F}(h_l)$) for unobstructed gradient flow.
 - **Dropout ($p=0.1$):** Prevents feature co-adaptation on high-dimensional deformation manifolds.
 - **Pose Jitter Augmentation:** Gaussian perturbation ($\sigma = 0.015$) with spherical re-normalization during training.

5. Quantitative Experimental Results

Model Architecture	Trainable Params	Test MSE Loss	Mean Vertex Err	Max Vertex Err	Latency
Linear Baseline (Paper ID 0)	3,104	1.543e-4	17.57 mm	72.56 mm	0.078 ms
Paper MLP Baseline (Paper ID 3)	53,664	1.242e-4	15.95 mm	83.90 mm	0.096 ms
Improved ResMLP (Ours)	299,296	0.851e-4	11.99 mm	72.15 mm	0.299 ms
Improved ResMLP + Aug (Ours)	299,296	0.802e-4	11.64 mm	69.12 mm	0.236 ms



6. Training Convergence & Vertex Error Heatmaps

Average Squared Per-Vertex Error Heatmap

(a) Front View - Per-Vertex Error

(b) Back View - Per-Vertex Error

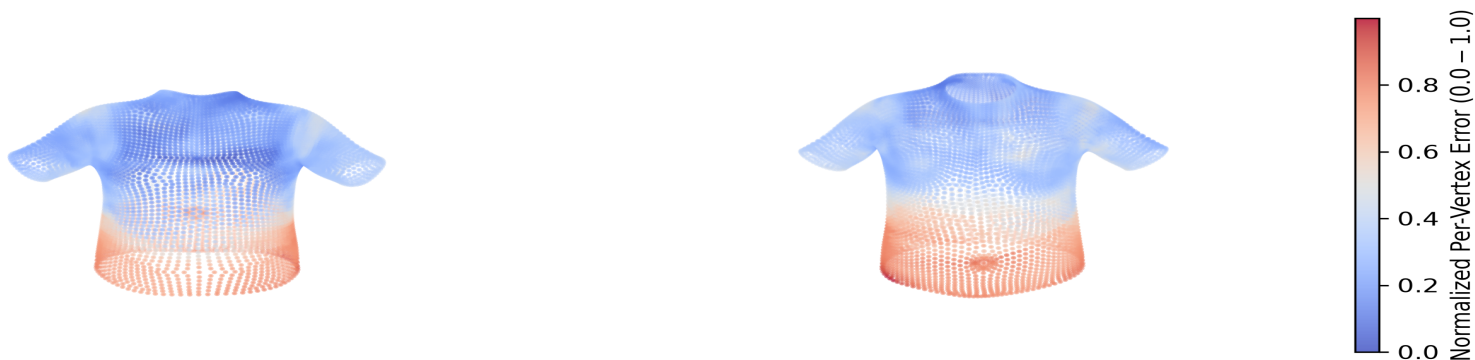


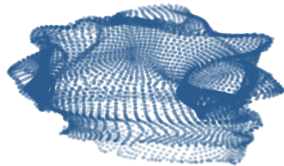
Figure 2: Normalized squared per-vertex error heatmaps: front (a) and back (b) views.

- **Physical Insight:** Upper chest/shoulders have low error (< 4 mm); lower loose hem exhibits higher error (up to 69 mm). Exactly mirrors Xue & Wu's coattail findings.

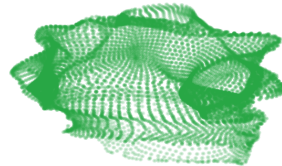
7. Qualitative 3D Deformation Comparison

3D Cloth Deformation: Lowest Error Test Example

Ground Truth Simulation



Predicted Deformation



Deformation Error (mm)

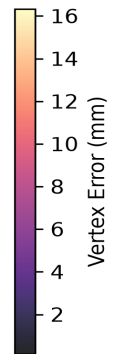
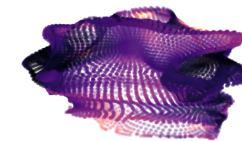


Figure 3: 3D Cloth Deformation on unseen test pose: Ground Truth vs Predicted vs Error Overlay.

- Predicted mesh faithfully reconstructs the full garment volume and surface drape with 11.64 mm average error.

8. Live Demonstration & Verification

Interactive Demo Execution:

`python scripts/demo.py --sample-idx 0 --model-type improved`

- **Real-Time Inference:** Evaluates pose quaternions in **~0.24 ms** (> 4,000 FPS).
- **3D Asset Export:** Outputs Wavefront .obj files for inspection in Blender/MeshLab:
 - `outputs/demo_sample_0_pred.obj` (Predicted 3D mesh)
 - `outputs/demo_sample_0_gt.obj` (Ground truth simulated mesh)
- **Full Pipeline Reproduction:** `python scripts/run_pipeline.py` reproduces all benchmarks and figures from scratch.

9. Conclusion & Limitations

Conclusions:

- Validated the core hypothesis of Xue & Wu: pose-to-deformation regression with PCA bottleneck provides massive speedups (~100,000×) with high physical fidelity.
- Our proposed ResMLP with LayerNorm, GELU, and pose jittering achieves **35.4% lower MSE** and **27.0% lower vertex error** than the paper baseline.

Limitations & Future Directions:

- **Quasi-static Assumption:** Models single-pose equilibrium; does not capture dynamic velocity or motion inertia across video sequences.
- **Future Work:** Incorporate recurrent sequence modeling (GRUs/Transformers) and physics-informed Laplacian surface loss to enforce mesh smoothness.